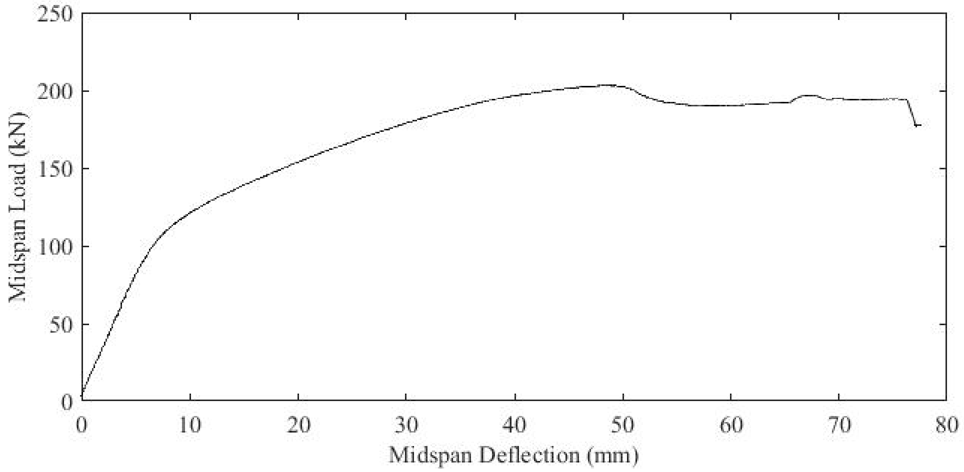
Analysis and Design of simple PC Beams - Ultimate limit State

Reference Text – Prestressed Concrete

Chapter 6 Flexure

THE UNIVERSITY OF QUEENSLAND
AUSTRALIA


MTS


Midspan Load (kN)
0
50
100
150
200
250
Midspan Deflection (mm)
0
10
20
30
40
50
60
70
80

Serviceability Limit State

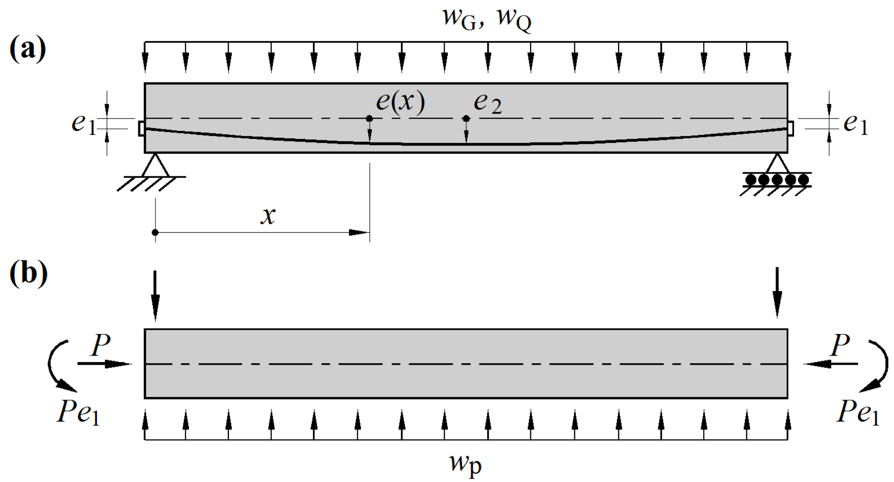
Short –term deflections *Uncracked* member

Fully **PC members, gross section properties are used**

Simply elastic formula for mid span deflection for simply support beam loaded with Uniformly distributed load

$$\Delta = \frac{5}{384}\frac{wL^4}{E_c I_g}$$

Initial Deflections



- Due to dead load w_G

$$\Delta_G = \frac{5}{384}\frac{w_G L^4}{E_c I_g}(down)\downarrow$$

- Due to live load w_Q

$$\Delta_Q = \frac{5}{384}\frac{w_Q L^4}{E_c I_g}(down)\downarrow$$

- Total short term deflection

$$\Delta_{GQp} = \Delta_G + \Delta_Q - \Delta_p$$

Short –term deflections *Cracked* member Cl8.5.3.1

Partially **prestressed concrete members**

A simplified method for calculating deflections in cracked prestressed members uses the ‘effective flexural stiffness $E_c I_{ef}$ (*given in AS 3600 Cl. 8.5.3*)

$$I_{ef} = I_{cr} + (I_g - I_{cr})\left(\frac{M_{cr}}{M^*_s}\right)^3 \leq I_{ef.\max}$$

The effective section moment of area I_{ef} is defined in AS3600 Cl 8.5.3.1 (c)

This is known as Branson's equation and requires cracked section analysis to calculate I_{cr}

M^*_s the maximum bending moment at the section based on short-term serviceability load

$$M_{cr} = Z(f'_{ct.f} - \sigma_{cs} + P / A_g) + Pe \geq 0$$

$I_{ef,max}$ = maximum effective second moment of area and is taken as I_g when $A_{st}/db >= 0.005$ or $0.6I_g$ when $A_{st}/bd < 0.005$

NOTE: If member is fully prestressed then $I_{ef}=I_g$

Long-term deflections

Additional long term deflection

$$\Delta_{LT} = \Delta_{pg} k_{cs}$$

k_{cs} (Cl. 8.5.3.2) and Δ_{pg} deflection due to applied prestress and self weight only (ie sustained loads)

$$k_{cs} = [2 - 1.2(A_{sc} / A_{st})] \geq 0.8$$

- Total long term deflection

$$\Delta_{total} = \Delta_{initial} + \Delta_{LT}$$

under short term service loads

$$\Delta_{initial} = \Delta_G + \psi_s \Delta_Q - \Delta_p$$

Deflections are then checked against suggested limits given in AS 3600 T2.3.2

Serviceability limit state design of PC members – crack control

Design at transfer P_i.

Only immediate losses have occurred. Member is subject to the minimum applied loading ie self-weight and prestress only.

Minimum allowable concrete strength at transfer = f_{cp}

The determines the upper limit of the magnitude of the prestressing force

AS3600 Clause 8.1.6.2 provides a ‘deemed to satisfy’ criterion if the maximum compressive stress in the concrete at transfer is limited to $0.5f_{cp}$

Serviceability limit state design of PC members – crack control (cont'd)

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Design for stress limit under full service load use P_e

Fully prestressed members are required to remain uncracked under full service load, and so maximum tensile stresses ≤ 0.25 $\sqrt{f'_c}$ Clause 8.6.2

For partially PC members, some cracking is permitted and Cl 8.6.2 requires:

reinforcement to be provided near the tensile face

the calculated maximum flexural tensile stress under short term service loads to ≤ 0.6 $\sqrt{f'_c}$ to control cracking in a 'deemed to satisfy' provision

Ultimate limit State

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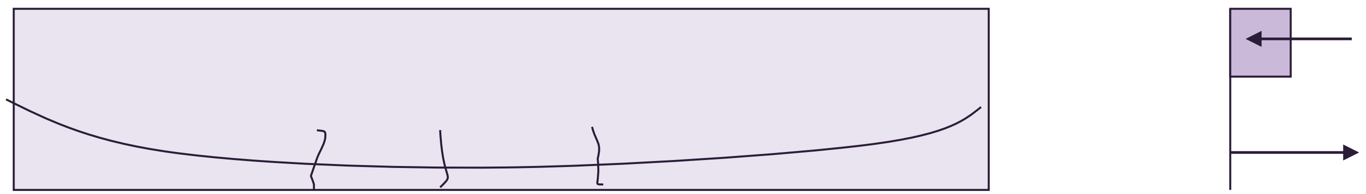
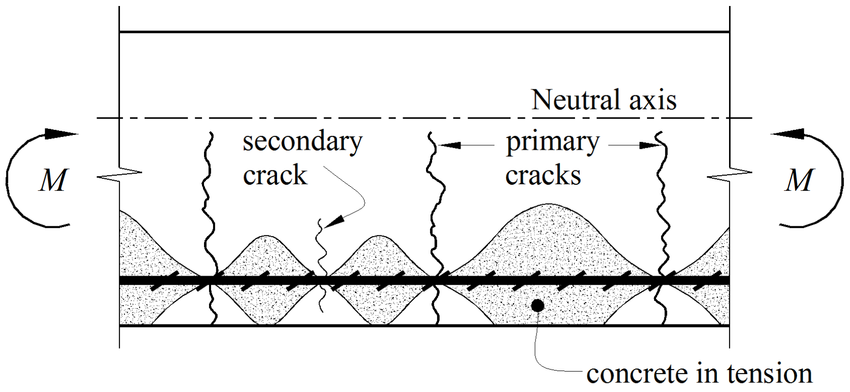
Flexural Strength of Prestressed Concrete Beams

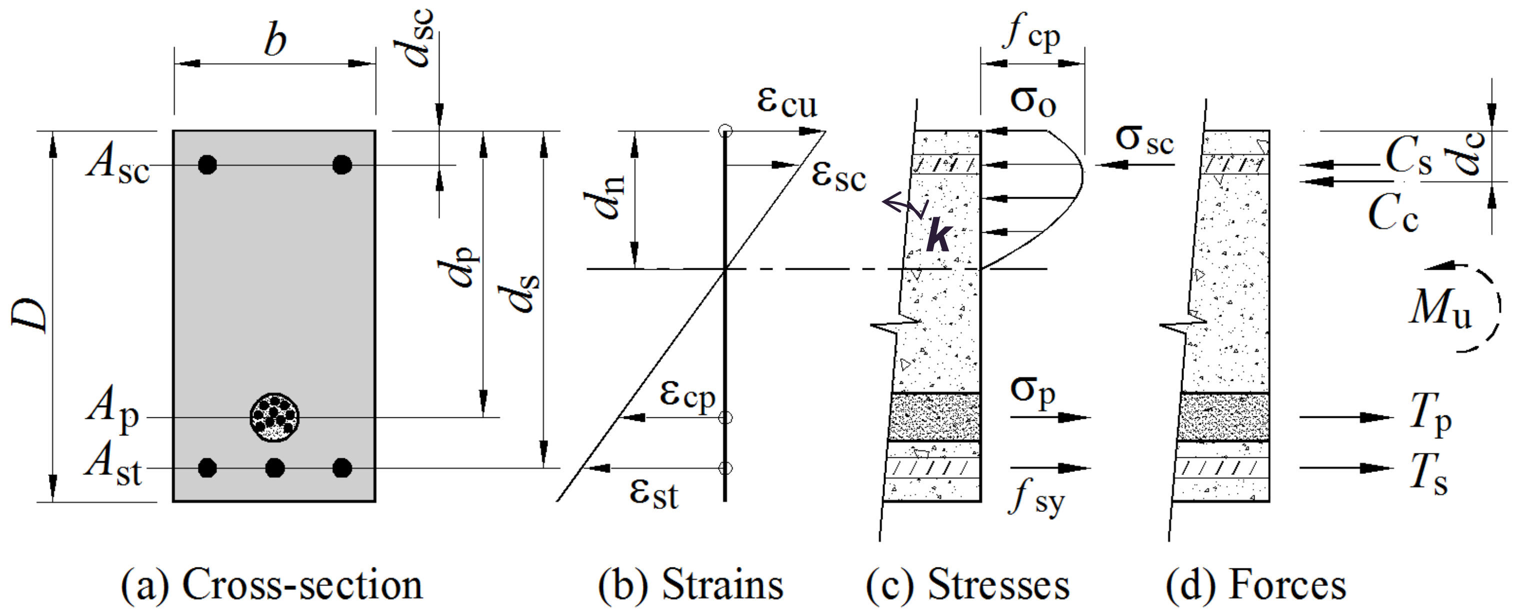
– Overload Behaviour





Conditions at M_u (Ultimate Moment) critical cross section
ie. Ultimate strength at overload

Sectional curvature $k = \varepsilon_o / d_n$ *(in this case* $\varepsilon_o = \varepsilon_{cu}$*)*



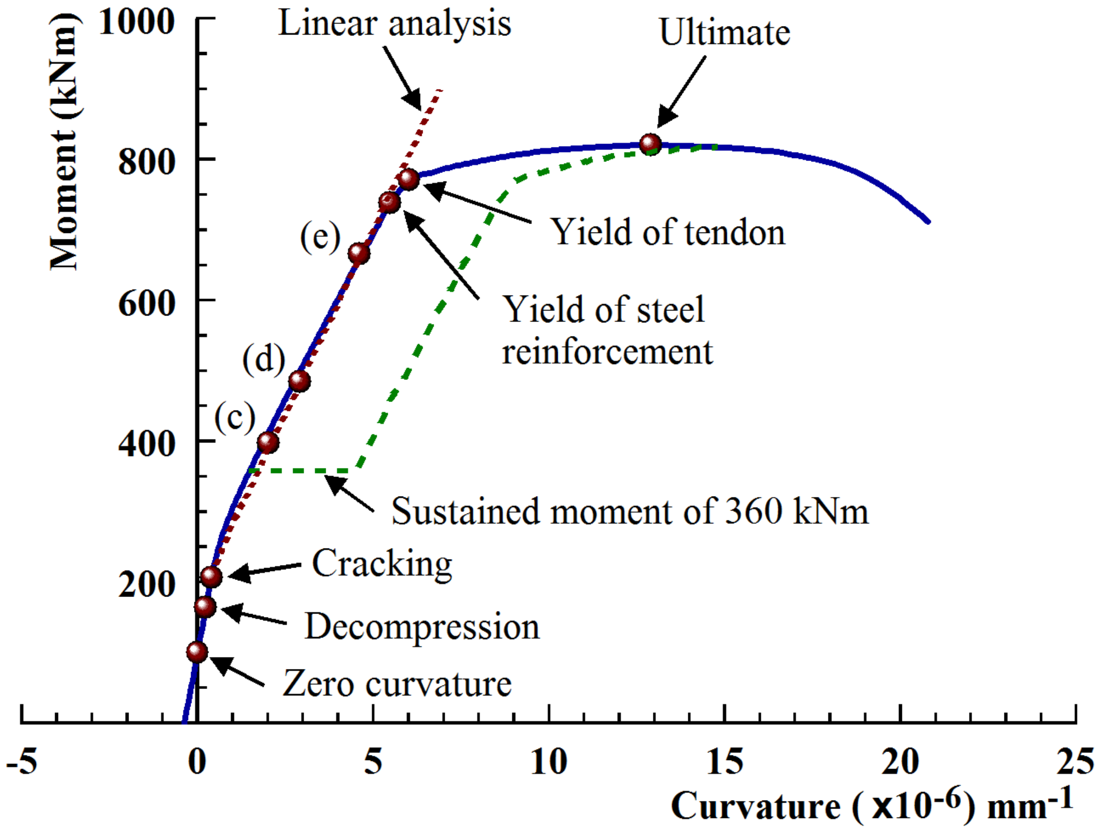
Equilibrium requirements

$C = T_s + T_p$

$M_u = T_s d_{st} + T_p d_p - C_c d_c - C_s d_{sc}$

Moment-Curvature relationship

- Moment curvature relationships provide a good indication of the overload behaviour of flexural members



Conditions at M_u (Ultimate Moment)
ie. Ultimate strength at overload

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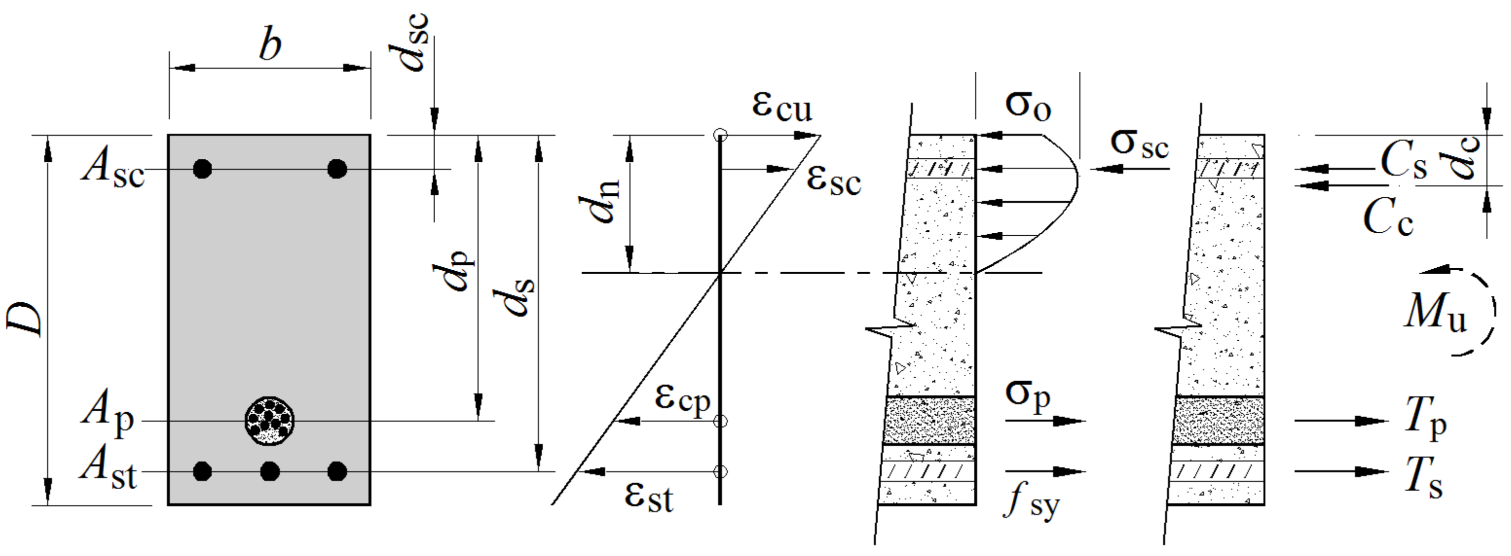


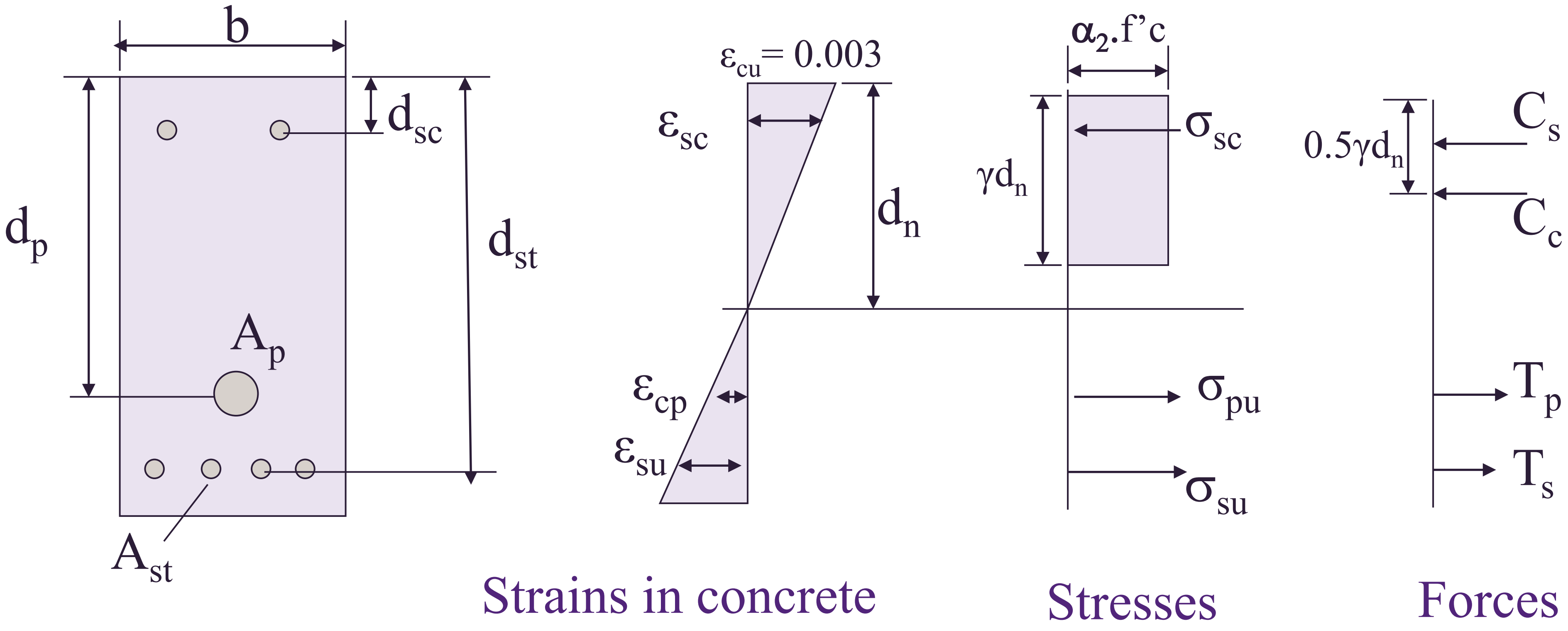
(a) Cross-section (b) Strains (c) Stresses (d) Forces

Equilibrium requirements

$C = T_s + T_p$

$M_u = T_s d_{st} + T_p d_p - C_c d_c - C_s d_{sc}$

Assumptions for Ultimate strength analysis



- Distribution of compressive stresses in concrete above neutral axis
- Magnitude of extreme fibre concrete strain ε_{cu}
- Distribution of strains throughout the depth of the section
- Stress-strain relationship for prestressing and reinforcing steel

(f_{py} refer to AS3600 Cl3.3.1, f_{sy} refer to AS3600 Cl3.2.1)

Ultimate strength of prestressed concrete members has to be checked at two points in time

At Transfer (AS3600 – 2018 cl.2.4.2)

Check whether the beam can sustain the applied prestress without failure

$\varphi\, P_u > 1.15\, Pj \qquad (\varphi = 0.6)$

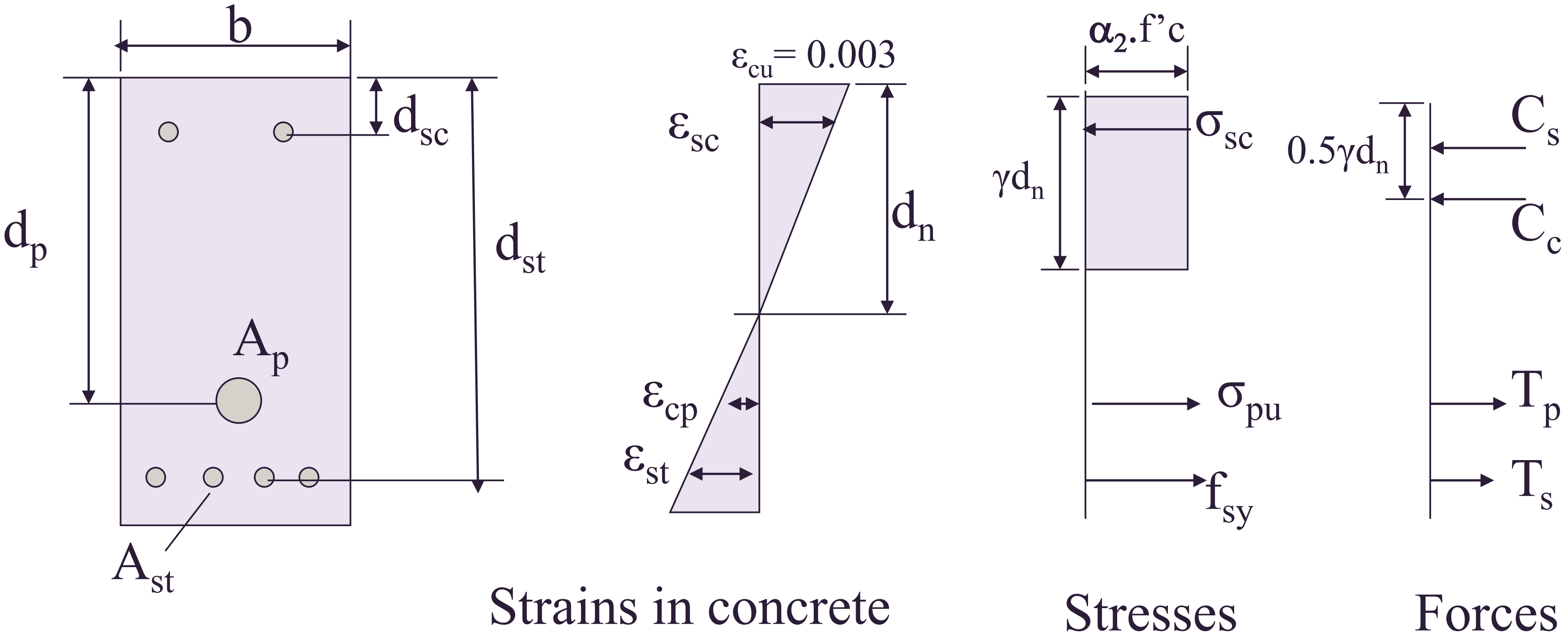
At Overload

Check whether the beam can sustain factored strength design loads without failure

$M^* \leq \varphi M_u$, $\varphi = 0.8$, $k_u \leq 0.4$

Since both checks are for failure possibility, these are ultimate strength checks.

Analysis of M_u using yield strength of prestressing steel
ie. Ultimate strength at overload



Note: rectangular stress block assumptions given in AS3600-2018 Clause 8.1.3

where $\alpha_2 = 0.85 – 0.0015f'c$ for $\alpha_2 \geq 0.67$

$\gamma = 0.97 – 0.0025f'c$ ($\gamma \geq 0.67$)

$d_n = k_u d$

d= effective depth

Calculation of forces in the section

Concrete Compressive force, C_c

$C_c = \alpha f'_c \gamma d_n b$ and acts at depth $d_c = 0.5\gamma d_n$

Steel compressive Force, C_{sc}

$C_s = A_{sc}E_s\varepsilon_{sc} = A_{sc}E_s \varepsilon_0 (d_n - d_{sc})/d_n$ (if compression steel has not yielded) or

$C_s = A_{sc}f_{sy}$ (if compression steel has yielded)

and acts at depth d_{sc}

Tensile forces on prestressing tendon T_p and reinforcing steel T_s

assuming both prestressing and tensile steel have

yielded,

$T_s = A_{st}f_{sy}$ and acts d_{st}

$T_p = A_p f_{py}$ and acts d_p

Calculation of M_u

Assumption all steels have yielded and neutral axis depth d_n determined from force equilibrium ie.

Force Equilibrium

$C_c = T_p + T_s - C_c$

$0.85f'_c \gamma d_n b = f_{sy}A_{st} + f_{py}A_p - f_{sy}A_{sc}$

With d_n evaluated and the top fibre strain fixed at ε_{cu} = 0.003, the strains in all the steel elements are determined by similar triangles

$$\varepsilon_{pu} = \varepsilon_{pe} + \varepsilon_{ce} + \varepsilon_{cu} \frac{d_p - dn}{d_n}$$

Calculation of M_u

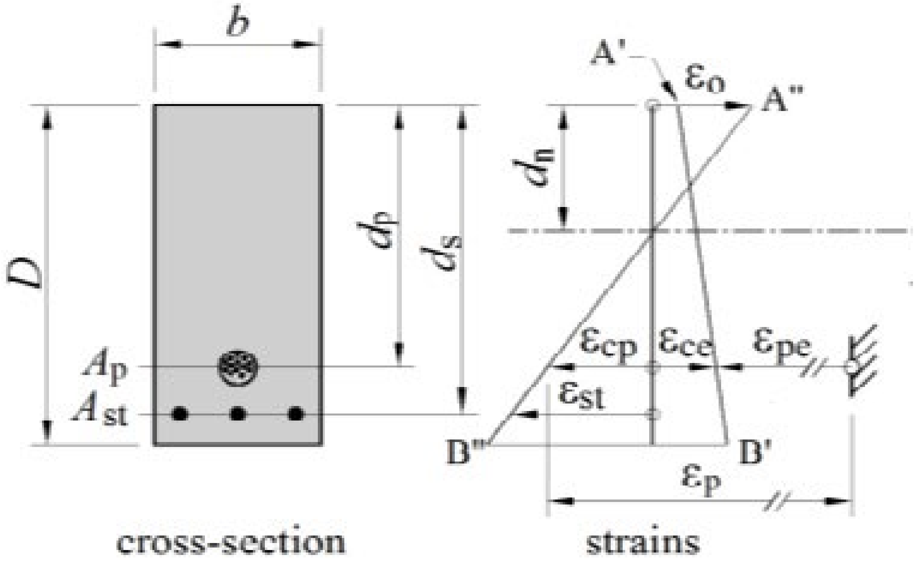
Provided all steel strains are greater than relevant limits ε_{sy} and ε_{py}, Moment Equilibrium,

Taking moments of the internal forces about the top fibre of the section,

$M_u = T_s d_{st} + T_p d_p - C_c d_c - C_s d_{sc}$

Where ***'Mu'*** is the ultimate moment capacity is the applied external moment corresponding to failure. Failure in a reinforced concrete beam is defined as the point at which maximum concrete compressive strain = *0.003*

Strains at M_u in section with prestress



Due to prestress only, strain in concrete at tendon level is ε_{ce} (+ve); strain in the tendon is ε_{pe} (-ve)

Applied moment causes strain in concrete to change from +ve ε_{ce} to –ve ε_{cp}. The concrete at the tendon level has therefore undergone a tensile strain increment $\Delta\varepsilon = \varepsilon_{cp} + \varepsilon_{ce}$ (absolute values). Assuming perfect bond, the same change in strain must occur in tendon.

Total strain in prestress $\varepsilon_p = \varepsilon_{pe} + \varepsilon_{ce} + \varepsilon_{cp}$

Total strain in prestressing steel ε_{pu}

ε_{pu} is the total strain in the prestressing steel at M_u.

Total strain in prestress steel = $\varepsilon_{pu} = \varepsilon_{pe} + \varepsilon_{ce} + \varepsilon_{cp}$

$$= \varepsilon_{pe} + \varepsilon_{ce} + \varepsilon_0(d_p - d_n)/d_n$$

due to effective prestress

$\varepsilon_{pe} = \sigma_p/E_p = P/A_pE_p$

Compressive stress in concrete adjacent to the prestressing cable, due to prestress alone

$\varepsilon_{ce} = \sigma_{ce}/E_c$ where $\sigma_{ce} = P/A_g + Pe^2/I_g$

Tensile strain in concrete at level of prestress steel due to applied Moment

$\varepsilon_{cp} = \varepsilon_0(d_p - d_n)/d_n$

Analysis with semi-empirical expressions for σ_{pu}

Cl. 8.1.7 of AS3600 permits use of a simple expression for σ_{pu} in bonded tendons at M_u.

$$\sigma_{pu} = f_{pb}\left[1 - \frac{k_1 k_2}{\gamma}\right]$$

In which $k_1 = 0.40$ if the ratio of the yield stress of the tendon, f_{py} , to the minimum breaking stress f_p is less than 0.9,

and $k_1 = 0.28$ if the ratio exceeds 0.9

$$k_2 = \frac{1}{b_{ef} d_p f_c'}(A_{pt} f_{pb} + (A_{st} - A_{sc}) f_{sy})$$

If compressive reinforcement is taken into account k_2 cannot be less than 0.17 and d_{sc} cannot be greater than 0.15 d_p

(Note: refer to Clauses 3.3.1 for tendon properties)

Analysis with semi-empirical expressions (cont'd)

This obviates the need to analyse strains.

Thus, for section with compression reinforcement also, assuming that tensile steel has yielded

$T = T_p + T_s = \sigma_{pu}A_p + A_{st}f_{sy}$ with σ_{pu} from above

$C = C_c + C_s = \alpha f'_c b\gamma d_n + A_{sc}\sigma_{sc}$

Initially assume that compression steel has also yielded,

Thus $\Sigma F_x=0$ gives us $C = T$ from which

$d_n = (A_p\,\sigma_{pu} + A_{st}f_{sy} - A_{sc}f_{sy})/(\alpha f'_c b\gamma)$

Analysis with semi-empirical expressions (cont'd)

Check that compression steel has yielded. If not, revise estimate of σ_{sc} and iterate to solution

Taking moments about tensile steel,

$M_u = C_c(d_{st}-0.5\gamma d_n) + C_s(d_{st}-d_{sc}) - \sigma_{pu}A_p(d_{st}-d_p)$

Must still carry out check for ductility

Addition of reinforcement to increase M_u

Normal design procedure

Prestressing force determined from service load requirements

Unstressed reinforcement added to satisfy the ultimate strength requirements

Approximate expression for determining ordinary reinforcing steel, provided reinforcing and prestressing steel at yield, where $M_u = M^*/\phi$

$$A_{st} = \frac{M_u - A_p f_{py}(d_p - 0.15 d_{st})}{f_{sy}(0.85 d_{st})}$$

Strength at Transfer of prestressed beam (section 6.9.5 text book)

Possible for prestressed beam to fail during transfer if force in the tendon becomes large enough to cause the concrete surrounding the cable to crush.

While beam is being prestressed, any cross-section is subjected to two actions: 1) the prestressing force P and 2) a bending moment M_G (beam self- weight plus any other loads acting at transfer)

The conditions of equilibrium at transfer require
-the sum of all forces must be zero
-the sum of the moments of all forces about any point must be equal to M_G

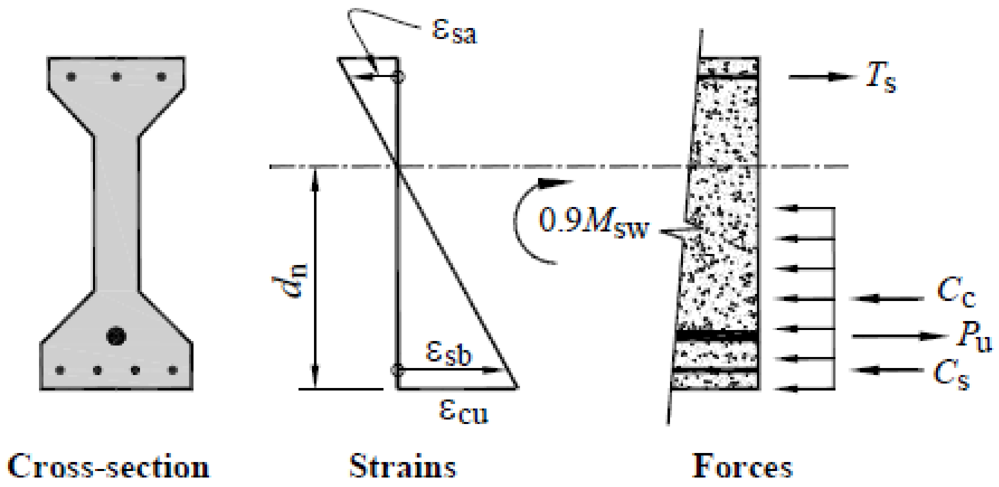
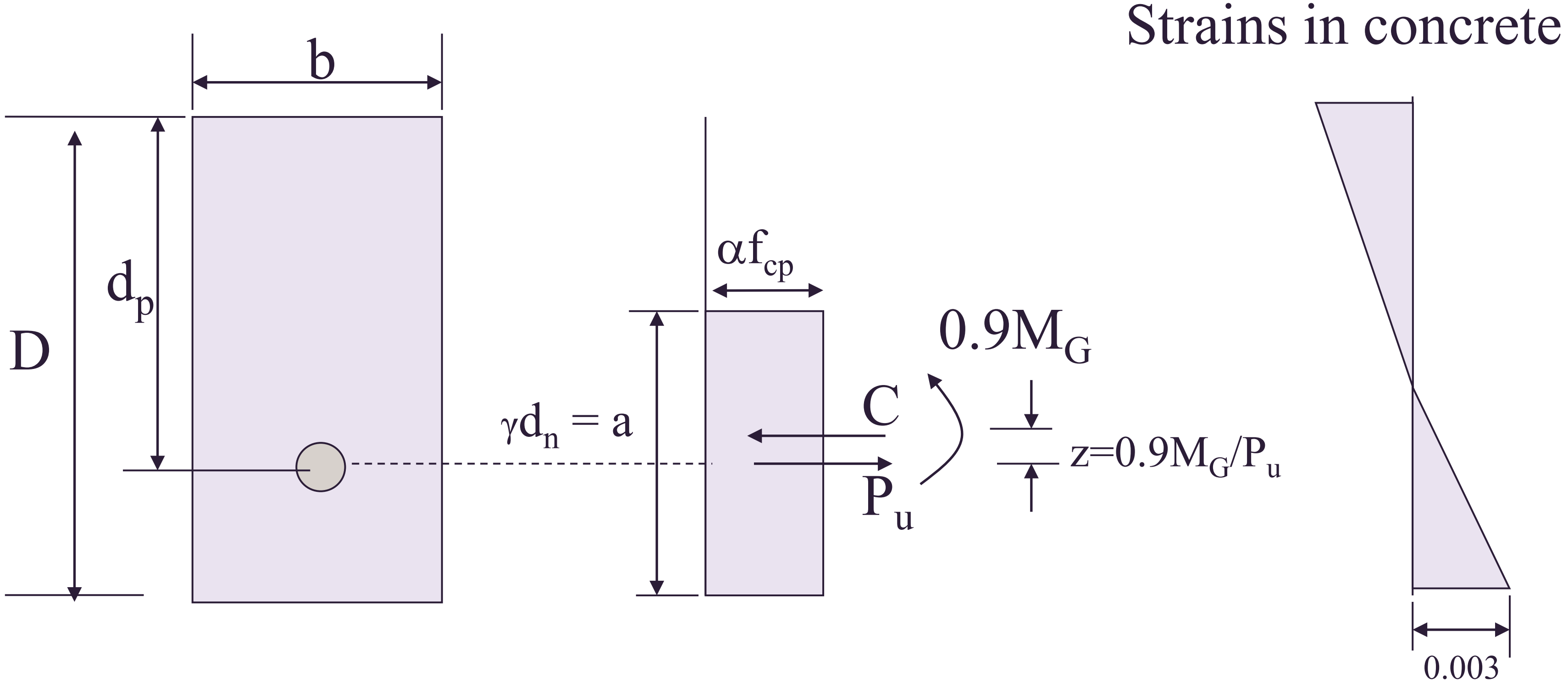


Figure 6.12 Conditions at transfer

Strength Check at Transfer



Applied external bending moment is the dead load moment: M_G

P_u = the prestress force which will create failure of the section (bottom concrete strain = 0.003)

C is the concrete compressive strength resultant acts at a distance a/2 from bottom of section, $C = \alpha f_{cp} ab$

Calculation of P_u

For horizontal force equilibrium, $C = P_u$

For moment equilibrium, $P_u z = M_G$

where z is the lever arm distance between C and P_u

$z = M_G/P_u$ M_G=self-weight moment x load factor

(ie load factor = 0.9 Ref.AS3600-2018 cl2.4.2)

Hence, $a = 2[(D-d_p) + M_G/P_u]$

$C = \alpha f_{cp} a.b$

Force equilibrium, $C = P_u$,

Solution of this quadratic equation gives P_u, the force at which the section would reach its strength limit state

<u>AS3600 Design check</u> (cl.2.4.2)

$\varphi P_u > 1.15 P_j$ where φ = 0.6 and 1.15 is the load factor

and a load factor of 0.9 on M_G,

AS3600 deemed to comply method for design at transfer

Cl. 8.1.4.2 states that strength at transfer is deemed to be satisfied if the maximum concrete compressive stress at transfer ≤ *0.5* f_{cp} *(stress distribution of rectangular) or 0.6* f_{cp} *(stress distribution of triangular)*

For sections without conventional longitudinal reinforcement it is usually found that this ‘deemed to comply’ provision is easier to satisfy ie less conservative than the more rigorous strength calculations

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Example: Calculation of ultimate moment capacity

The University of Queensland
Australia

A post-tensioned prestressed concrete beam has a configuration of reinforcement shown in the figure. Calculate the ultimate moment capacity of the section.

f'_c = 32 MPa

γ = 0.89, α 0.80

E_c = 28,570 MPa

E_p = 190,000 MPa

A_{st} = 1800 mm^2

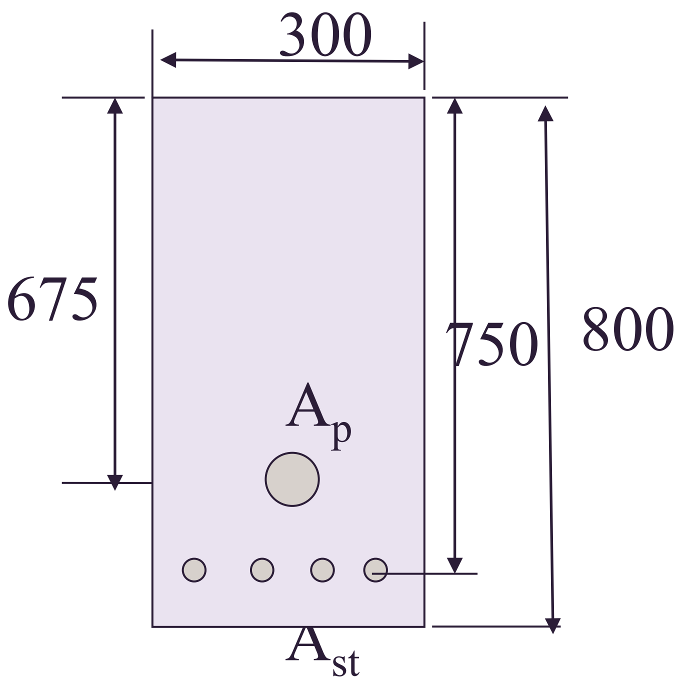
f_{sy} = 400 MPa

f_{py} = 1600 MPa

A_p = 500 mm^2

ε_{ce} = 0.000176

σ_{pe} = 1000 MPa



Example – strength at transfer

A simply supported post-tensioned beam spans 25 m and has a rectangular cross section shown in the figure. Concrete strength at transfer = 32 MPa. Calculate the magnitude of the prestress force P_u at which the section would fail under crushing of concrete at transfer.

A_{st} = 3000 mm^2

Self-weight = 0.35 x 1.2 x 25 = 10.5 kN/m

M_G = 10.5 x 25^2/8 = 820 kNm

γ = 0.89, α 0.80